

L'Hôpital's Rule

$$\lim_{x \rightarrow 1} \frac{e^x - 1}{x}$$

$$\Rightarrow \frac{e^0 - 1}{0} = \frac{1 - 1}{0} = \frac{0}{0} \quad \underline{\underline{\text{I.F}}}$$

H.R:

$$\frac{e^x}{1} = \frac{1}{1} = 1$$

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

$$\Rightarrow \frac{e^{2(0)} - 1 - 2(0)}{(0)^2} = \frac{1 - 1}{0} = \frac{0}{0} \quad \underline{\underline{\text{I.F}}}$$

H.R:

$$\frac{2e^x - 2}{2x} = \frac{2e^0 - 2}{2(0)} = \frac{0}{0}$$

$$\Rightarrow \left(\frac{2e^x - 2}{2x} \right)' = \frac{2e^x}{2} = e^x = e^0 = 1$$

